Neurodevelopmental Disorder with Microcephaly, Movement Abnormalities, and Seizures (NEDMIMS): A Comprehensive Disease Characterization

Disease: Neurodevelopmental Disorder with Microcephaly, Movement Abnormalities, and Seizures (NEDMIMS) **OMIM:** #620023 · **MONDO:** MONDO:0859282 · **UMLS:** C5774208 · **Causal gene:** *CHKA* (choline kinase alpha, 11q13.2) · **Category:** Mendelian (autosomal recessive)

> **Evidence caveat.** This is an ultra-rare, recently delineated Mendelian disorder. Almost all *disease-specific* clinical statements derive from a single primary report of six individuals from five families (PMID: 35202461). Mechanistic, model-organism, and pathway statements draw on broader choline-kinase / Kennedy-pathway literature. Where information does not exist, this is stated explicitly.

Summary

Neurodevelopmental Disorder with Microcephaly, Movement Abnormalities, and Seizures (**NEDMIMS**, OMIM **#620023**) is an ultra-rare, autosomal recessive Mendelian disorder caused by **biallelic (homozygous or compound heterozygous) loss-of-function/hypomorphic variants in *CHKA***, the gene encoding **choline kinase alpha** on chromosome 11q13.2. The disease was first delineated by Klöckner and colleagues in 2022, who reported six affected individuals from five unrelated families (PMID: 35202461). CHKA catalyzes the **first committed step of the CDP-choline (Kennedy) pathway** — the ATP-dependent phosphorylation of choline to phosphocholine — the rate-influencing entry point for de novo synthesis of **phosphatidylcholine (PC)**, the most abundant phospholipid of eukaryotic cell membranes. Disease variants reduce CHKA enzymatic activity, starving the developing brain of membrane phospholipid supply.

The clinical phenotype is severe and highly stereotyped: **fully penetrant microcephaly, severe-to-profound global developmental delay with absent speech, early-onset refractory epileptic encephalopathy, and a mixed movement disorder** combining spasticity/hypertonia with hyperkinetic features (dyskinesia, choreoathetosis). Additional features include hyperreflexia, inability to walk, short stature, nystagmus, aggressive and self-injurious behavior, scoliosis, and — on neuroimaging — CNS hypomyelination and thin corpus callosum. Onset is congenital-to-infantile. There is no biochemical diagnostic marker; diagnosis relies on **exome or genome sequencing**. Management is entirely **supportive** (anti-seizure medications, spasticity/movement management, physical/occupational/speech therapy, orthopedic and feeding support); no disease-specific or curative therapy exists.

Mechanistically, NEDMIMS joins a growing family of "**Kennedy-pathway disorders**" (alongside *CHKB*, *PCYT1A*, *PCYT2*, and *SELENOI* diseases). Because complete *Chka* knockout is embryonic lethal in mice, human disease alleles are necessarily **hypomorphic** (partial loss of function), not null. Functional validation in yeast complementation assays and structural modeling confirmed reduced catalytic activity of disease variants. The successful AAV-mediated gene-therapy rescue of the paralogous *Chkb* (rmd) mouse suggests phospholipid-pathway defects are therapeutically tractable in principle, offering a rationale for future disease-modifying strategies.

Key Findings

Finding 1 — Disease identity: NEDMIMS is caused by biallelic *CHKA* variants

NEDMIMS (OMIM #**620023**) was established by Klöckner et al. (2022, *Brain*), who identified **six individuals from five unrelated families** carrying bi-allelic variants in *CHKA* (gene MIM 118491; 11q13.2). Inheritance is **autosomal recessive**. The authors state: *"We identified six individuals from five families with bi-allelic variants in CHKA presenting with severe global developmental delay, epilepsy, movement disorders and microcephaly"* (PMID: 35202461). The four cardinal features give the disorder its name. Yeast complementation and structural modeling showed the variants *"reduce the enzymatic activity of CHKA and confer a significant impairment of the first enzymatic step of the Kennedy pathway,"* establishing a **loss-of-function mechanism**.

Finding 2 — CHKA loss cripples the CDP-choline (Kennedy) pathway

Choline kinase (isoforms CHKα/CHKβ) catalyzes **ATP + choline → ADP + phosphocholine**, the first committed step of de novo PC synthesis; phosphocholine is converted to **CDP-choline** (PCYT1A/B) and then to **phosphatidylcholine**. Klöckner et al. note: *"The Kennedy pathways catalyse the de novo synthesis of phosphatidylcholine and phosphatidylethanolamine, the most abundant components of eukaryotic cell membranes"* (PMID: 35202461). This is corroborated by Chen et al.: *"Choline kinases possess enzyme activity that catalyses the conversion of choline to phosphocholine, which is further converted to cytidine diphosphate-coline (CDP-choline) in the biosynthesis of phosphatidylcholine (PC)"* (PMID: 27769579). **Four of ten Kennedy-pathway genes** were already disease-associated before NEDMIMS: *CHKB* (megaconial muscular dystrophy), *PCYT1A* (cone-rod dystrophy/bone), *PCYT2* and *SELENOI* (spastic paraplegia).

Finding 3 — *CHKA* is not haploinsufficient, consistent with recessive disease

gnomAD constraint for *CHKA* (ENSG00000110721): **pLI ≈ $4.3{\times}10^{-6}$**, observed/expected LoF = **0.57 (90% CI 0.43–0.75; LOEUF 0.75)**, LoF Z = 2.86, missense Z ≈ 0. Heterozygous loss-of-function is **tolerated** in the general population (carriers unaffected); disease requires **biallelic** variants — fully consistent with autosomal recessive inheritance and the disorder's ultra-rarity/consanguinity association.

Finding 4 — Complete Chka loss is embryonic lethal; disease alleles are hypomorphic

Wu et al.: *"Disruption of murine Chka leads to embryonic lethality, whereas a spontaneous genomic deletion in murine Chkb results in neonatal forelimb bone deformity and hindlimb muscular dystrophy"* (PMID: 20026284). Because a null *Chka* genotype is incompatible with life in mice, viable **human disease alleles must retain residual activity (hypomorphic)**. The paralogous disorder offers a therapeutic proof-of-concept: *"Intramuscular gene therapy post-disease onset using an adeno-associated viral 6 (AAV6) vector carrying a functional copy of Chkb is also capable of rescuing the dystrophy phenotype"* (PMID: 31216357). Human variants were additionally validated by **yeast (S. cerevisiae) complementation**.

Finding 5 — Full phenotype spectrum with HPO frequencies (n=6, Klöckner 2022)

Fully penetrant core features (6/6, 100%): microcephaly (HP:0000252), severe global developmental delay (HP:0011344), delayed gross motor development (HP:0002194), absent speech (HP:0001344), epileptic encephalopathy (HP:0200134), hypertonia (HP:0001276). Highly frequent: hyperreflexia (HP:0001347, 5/5), inability to walk (HP:0002540, 5/6). Common: infantile onset (HP:0003593, 4/6), short stature (HP:0004322, 4/6), nystagmus (HP:0000639, 3/6), aggressive behavior (HP:0000718, 3/6), scoliosis (HP:0002650, 3/6). Movement axis: dyskinesia (HP:0100660, 2/6), choreoathetosis (HP:0001266, 1/6), rigidity (HP:0002063, 1/6). Less frequent (1/6): self-injury (HP:0100716), sleep disturbance (HP:0002360), autistic behavior (HP:0000729), hyperactivity (HP:0000752), cerebral visual impairment (HP:0100704), feeding difficulties (HP:0011968), high palate (HP:0000218), kidney stone (HP:0000787). Neuroimaging: CNS hypomyelination (HP:0003429, 2/5), thin corpus callosum (HP:0033725, 1/5). Source: *"...severe global developmental delay, epilepsy, movement disorders and microcephaly"* (PMID: 35202461).

Finding 6 — *CHKA* pathogenic variant spectrum (ClinVar; NM_001277.3)

Pathogenic/Likely-pathogenic NEDMIMS variants span classes: **c.1030G>T p.(Gly344Ter)** (nonsense); **c.14dup p.(Cys6fs)** (frameshift); **c.1021T>C p.(Phe341Leu)**, **c.580C>T p.(Pro194Ser)**, **c.421C>T p.(Arg141Trp)** (missense). Full *CHKA* ClinVar record (94 entries): **7 Pathogenic, 10 Likely pathogenic, 54 VUS, 5 Likely benign** — though many "Pathogenic" entries are large 11q contiguous-gene CNVs, not NEDMIMS-specific. Variants are **germline, biallelic**; consequence is **loss-of-function / reduced choline kinase activity (hypomorphic)**.

Finding 7 — CHKA protein: cytosolic choline/ethanolamine kinase

CHKA (UniProt **P35790**; HGNC:1937; NCBIGene:1119; ENSG00000110721; 11q13.2). Reactions: choline + ATP → phosphocholine + ADP (**GO:0004103**) and ethanolamine + ATP → phosphoethanolamine + ADP (**GO:0004305**). Processes: PC biosynthesis (**GO:0006656**), CDP-choline pathway (**GO:0006657**), PE biosynthesis (**GO:0006646**). Localization: cytosol (**GO:0005829**), cytoplasm (**GO:0005737**), lipid droplet (**GO:0005811**). CHEBI entities: choline (CHEBI:15354), ATP (CHEBI:30616), phosphocholine (CHEBI:295975), CDP-choline (CHEBI:16436), phosphatidylcholine (CHEBI:16110), ethanolamine (CHEBI:57603).

Finding 8 — Orthologs and pathway memberships

CHKA (HomoloGene 88575) orthologs: mouse *Chka* (12660), rat *Chka* (29194), zebrafish *chka* (558499), *C. elegans* (180703), plus dog/cow/chicken/chimp. Pathways: KEGG **hsa00564** (glycerophospholipid metabolism), **hsa05231** (choline metabolism in cancer); WikiPathways **WP3933** (Kennedy pathway); Reactome **R-HSA-1483191** (synthesis of PC). No naturally occurring *CHKA*-equivalent disease is catalogued in OMIA for domestic animals.

Finding 9 — Rich experimental structural data for variant mapping

Human CHKα (457 aa) has **≥5 high-resolution PDB structures** (2IG7, 2CKQ, 2CKO, 3F2R, 2CKP). Architecture: **disordered N-terminal region (~res 1–86)** + **choline/ethanolamine kinase catalytic domain (~res 80–457)**; ATP/substrate-contacting residues include 117–123, 119–121, 146, 207–213, 308, 330. NEDMIMS missense variants (**p.Arg141Trp, p.Pro194Ser, p.Phe341Leu**) fall **within/adjacent to the catalytic domain**, structurally consistent with the measured reduction in activity (PMID: 35202461).

Mechanistic Model / Interpretation

Causal chain

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Biallelic hypomorphic CHKA variants (missense / nonsense / frameshift)
        │  (null is embryonic-lethal → alleles retain residual activity)
        ▼
Reduced choline kinase activity
        │  ATP + choline ──X──► phosphocholine   (first committed step)
        ▼
Impaired CDP-choline (Kennedy) pathway flux
        ▼
Deficient de novo phosphatidylcholine (± phosphatidylethanolamine) synthesis
        │  PC = most abundant membrane phospholipid
        ▼
Inadequate membrane phospholipid supply in the developing CNS
        │  (neuronal membrane biogenesis, myelination, synaptogenesis)
        ▼
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│ Microcephaly   │ Hypomyelination/thin CC │ Epileptic          │
│ (HP:0000252)   │ (HP:0003429/HP:0033725) │ encephalopathy     │
│                │                         │ (HP:0200134)       │
│ Severe global DD / absent speech (HP:0011344 / HP:0001344)   │
│ Mixed movement disorder: hypertonia + dyskinesia/chorea      │
│ (HP:0001276 / HP:0100660 / HP:0001266)                       │
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Upstream vs downstream. The upstream trigger is the genetic lesion reducing CHKA catalytic output; the immediate downstream node is diminished phosphocholine production and reduced Kennedy-pathway flux to PC. Distal clinical consequences (microcephaly, hypomyelination, seizures, movement disorder) reflect the developing brain's exceptional dependence on continuous membrane phospholipid biosynthesis for neuronal proliferation, axonal/dendritic membrane expansion, and myelin production. CHKA's bifunctional ethanolamine kinase activity (GO:0004305) means PE synthesis may also be partly compromised, compounding the deficit.

Placement within the Kennedy-pathway disorder family

Gene	Enzyme / step	Associated disorder	Relationship to NEDMIMS
CHKA	Choline kinase α (step 1, choline branch)	**NEDMIMS (OMIM #620023)**	Index disorder
CHKB	Choline kinase β (step 1, muscle-predominant)	Megaconial congenital muscular dystrophy	Shares ID/DD/behavioral features; muscle-predominant, megaconial mitochondria (PMID: 23945283)
PCYT1A	CTP:phosphocholine cytidylyltransferase (step 2)	Cone-rod dystrophy, skeletal disease	Downstream, same pathway
PCYT2	Ethanolamine-branch cytidylyltransferase	Hereditary spastic paraplegia	Spasticity overlap
SELENOI	Ethanolaminephosphotransferase	Hereditary spastic paraplegia	Spasticity overlap

The **tissue-specific expression of the two paralogs** (Wu et al., PMID: 20026284) explains why α-isoform loss manifests as a neurodevelopmental (rather than myopathic) disease. Cell types: **neurons (CL:0000540)**, **oligodendrocytes (CL:0000128**; hypomyelination), and neural progenitors (microcephaly). Subcellular convergence: cytosolic phosphorylation (GO:0005829) → ER-based PC assembly → deficient plasma-membrane / myelin phospholipid.

Section-by-Section Disease Characterization

1. Disease Information

- **Overview:** ultra-rare autosomal recessive Mendelian neurodevelopmental disorder defined by microcephaly, severe global developmental delay, epileptic encephalopathy, and a mixed movement disorder, caused by biallelic *CHKA* variants that impair phosphatidylcholine biosynthesis.

- **Identifiers:** OMIM #**620023**; gene OMIM 118491; MONDO **:0859282**; UMLS **C5774208**; MedGen linked; HGNC:1937; NCBIGene:1119; Ensembl ENSG00000110721; UniProt P35790. **No dedicated Orphanet, ICD-10/ICD-11, or MeSH code** was identified (maps to broad categories: ICD-10 Q02 microcephaly / G40 epilepsy / F79 intellectual disability).
- **Synonyms:** NEDMIMS; CHKA-related neurodevelopmental disorder; CHKA-related developmental and epileptic encephalopathy with microcephaly; choline kinase alpha deficiency (neurodevelopmental form).
- **Source type:** aggregated disease-level knowledge from an individual-patient case series (n=6), not EHR/registry data.

2. Etiology

- **Causal factor:** purely **genetic** — biallelic hypomorphic loss-of-function variants in *CHKA*; no environmental/infectious cause. Klöckner et al. showed variants *"reduce the enzymatic activity of CHKA"* (PMID: 35202461).
- **Genetic risk factors:** inheriting two damaging *CHKA* alleles; **consanguinity/founder homozygosity** is the dominant risk context. *CHKA* is not haploinsufficient (pLI≈0), so carriers are unaffected. No proven modifier genes (paralog *CHKB* and downstream *PCYT1A/PCYT2/SELENOI/CHPT1/CEPT1* are plausible but unproven modifiers).
- **Environmental/protective factors:** none established. **Dietary choline** is a theoretical (untested) substrate-supply modifier.
- **Gene–environment interactions:** none documented.

3. Phenotypes

See Finding 5 for the HPO-annotated frequency profile. Phenotype types span clinical signs (microcephaly, hypertonia, hyperreflexia), developmental deficits (global DD, absent speech, inability to walk), behavioral changes (aggression, self-injury, autistic behavior, hyperactivity, sleep disturbance), and neuroimaging abnormalities (hypomyelination, thin corpus callosum). **Onset:** infantile (4/6) to childhood (2/6). **Severity:** severe-to-profound. **Progression:** developmental delay static-to-slowly-progressive; microcephaly progressive/postnatal; epilepsy often refractory. **Laboratory:** no diagnostic biochemical marker. **QoL impact:** profound — non-verbal, non-ambulatory in most, lifelong dependent care; no disease-specific QoL instrument data.

4. Genetic / Molecular Information

- **Causal gene:** *CHKA* (HGNC:1937; MIM 118491; 11q13.2; NM_001277.3 / NP_001268.1). See Finding 6 for variants.
- **Classification landscape:** 94 ClinVar records (7 P, 10 LP, 54 VUS, 5 LB); ACMG/AMP-based, supported by functional data + rarity + recessive segregation.
- **Allele frequency:** ultra-rare/absent in gnomAD; heterozygous LoF tolerated.
- **Origin:** germline, biallelic (CHKA somatic overactivity is a *cancer* phenomenon, not relevant here).
- **Functional consequence:** loss-of-function/hypomorphic; not gain-of-function or dominant-negative.
- **Modifier/epigenetic/chromosomal:** none NEDMIMS-specific; large 11q13 CNVs cause distinct contiguous-gene phenotypes.

5. Environmental Information

Not applicable — no toxic, lifestyle, or infectious contributors. (Diagnostic pitfall: intracranial calcifications in the broader spectrum can mimic congenital TORCH infection.)

6. Mechanism / Pathophysiology

Detailed above. Pathway: **CDP-choline / Kennedy pathway** (WP3933; KEGG hsa00564; Reactome R-HSA-1483191). Core defect: reduced **choline kinase activity (GO:0004103)** → impaired **PC biosynthesis (GO:0006656)**. Protein dysfunction: hypomorphic activity reduction with disease missense residues mapping to the catalytic domain (PDB 2IG7, 2CKQ, etc.). CHKA also has a non-catalytic scaffolding role (binds c-Src SH3 via its poly-proline region, PMID: 31745227); relevance to NEDMIMS unproven. No immune, infectious, or storage component. **No patient-level omics** (transcriptomic/proteomic/lipidomic) published; predicted lipidomic signature = reduced PC species.

7. Anatomical Structures Affected

- **Organ/system:** brain/CNS (UBERON:0000955, UBERON:0001017) — cortex (UBERON:0000956), white matter (UBERON:0002316), corpus callosum (UBERON:0002336); secondary skeletal (scoliosis; UBERON:0001130), growth, occasional renal.
- **Tissue/cell:** nervous tissue — neurons (CL:0000540), oligodendrocytes (CL:0000128; hypomyelination); no primary muscle pathology (contrast *CHKB*).

- **Subcellular:** cytosol (GO:0005829), lipid droplet (GO:0005811); pathology converges on cellular membranes (ER GO:0005783, plasma/myelin membrane).
- **Localization:** bilateral, symmetric, diffuse; not focal.

8. Temporal Development

- **Onset:** congenital-to-infantile (HP:0003593 4/6; childhood HP:0011463 2/6); chronic/insidious-developmental, with epilepsy sometimes acute in presentation.
- **Progression:** static-to-slowly-progressive developmental deficits; progressive microcephaly; often refractory epileptic encephalopathy; no discrete stages.
- **Course/duration:** chronic, lifelong, non-remitting; no spontaneous remission of core disability.
- **Critical period:** prenatal–early-postnatal brain growth/myelination window — the theoretical window for metabolic intervention.

9. Inheritance and Population

- **Inheritance:** autosomal recessive (HP:0000007); biallelic (homozygous or compound heterozygous).
- **Penetrance:** complete for the core tetrad in reported biallelic cases (small denominator). **Expressivity:** variable for secondary features.
- **Anticipation/mosaicism:** not applicable (not a repeat disorder).
- **Founder/consanguinity:** consanguinity central; likely private/founder alleles per family rather than one global founder.
- **Carrier frequency:** not formally estimated; expected very low.
- **Epidemiology:** ultra-rare; prevalence/incidence not established (<~10 individuals reported; 6 in the defining paper). **Demographics:** reported families of Middle Eastern/North African/South Asian background (ascertainment via homozygosity); no sex bias expected (autosomal); pediatric diagnosis.

10. Diagnostics

- **Genetic testing is definitive:** trio **WES/WGS** or a neurodevelopmental/epilepsy/microcephaly gene panel including *CHKA*; confirm biallelic status + segregation. **GeneMatcher** enabled cohort assembly. CMA/karyotype/FISH help exclude 11q CNVs but are typically **normal** in NEDMIMS. mtDNA/repeat testing not indicated.

- **No biochemical biomarker;** routine metabolic screens unrevealing.
- **Imaging:** brain MRI may be normal or show hypomyelination/leukoencephalopathy, thin corpus callosum, ventriculomegaly, cortical abnormalities (± calcifications on CT). **EEG:** epileptiform/encephalopathic patterns. Serial head-circumference documents progressive microcephaly.
- **Differential diagnosis:** other Kennedy-pathway disorders (*CHKB* megaconial dystrophy — muscle-predominant, megaconial mitochondria; *PCYT2/SELENOI* HSP; *PCYT1A* retinal/skeletal), *PPFIBP1*-related disorder (microcephaly + periventricular calcifications), primary microcephaly (MCPH) genes, congenital TORCH infections, other early-infantile DEEs, hypomyelinating leukodystrophies.
- **Screening:** not part of newborn screening; carrier/cascade and prenatal/PGT-M feasible for known biallelic families.

11. Outcome / Prognosis

- **Function:** severe-to-profound; most non-ambulatory (inability to walk 5/6) and non-verbal (absent speech 6/6) with refractory epilepsy → lifelong dependence (ICF: severe limitations in mobility, communication, self-care).
- **Survival/mortality:** not systematically quantified; early-onset refractory epileptic encephalopathy carries elevated risk (including generic SUDEP risk). Long-term life expectancy undetermined.
- **Complications:** refractory seizures, aspiration/feeding difficulties, failure to thrive/short stature, scoliosis, contractures, behavioral/self-injury and sleep problems, cerebral visual impairment, rare nephrolithiasis.
- **Recovery potential:** none for core deficit (irreversible developmental involvement).
- **Prognostic factors:** presumed correlation of residual CHKA activity (allele severity) and seizure control with outcome — plausible, not statistically established. No validated prognostic biomarker.

12. Treatment

No disease-specific or curative therapy. Management is symptomatic, supportive, multidisciplinary. - **Pharmacotherapy:** anti-seizure medications (NCIT: Anticonvulsant Agent); spasticity/movement agents (baclofen, trihexyphenidyl, focal botulinum toxin); behavioral/sleep management. No CHKA-specific pharmacogenomics. - **Advanced/experimental (conceptual only): choline/CDP-choline (citicoline)** substrate supplementation — precedent in the paralogous *CHKB* disorder (Chen et al. propose CDP-choline supplementation may be beneficial, PMID: 27769579) — efficacy in CHKA **unproven**. **Gene replacement** — AAV6-*Chkb* rescued the rmd mouse (PMID: 31216357), a template for future CHKA work (CNS delivery/timing are hurdles). CHKA **inhibitor** trials exist but are oncology programs (opposite direction — irrelevant to this LoF disorder). - **Surgical/interventional:** epilepsy surgery generally not applicable (diffuse/genetic); orthopedic scoliosis/contracture management; gastrostomy as needed. - **Supportive/rehabilitative (mainstay):** physical, occupational, and speech/communication therapy; nutrition; vision/orthopedic care; caregiver support (NCIT: Rehabilitation Therapy).

13. Prevention

- **Primary:** reproductive genetics only — **genetic counseling** for consanguineous/at-risk couples, **carrier screening**, and, for couples with a prior affected child, **prenatal diagnosis** or **PGT-M**.
- **Secondary:** early molecular diagnosis → early seizure control and early developmental/rehabilitation intervention within the critical window.
- **Tertiary:** complication prevention (seizure management, aspiration precautions/nutrition, scoliosis/contracture prevention, behavioral/sleep support).
- **Immunization/public-health/environmental/prophylaxis:** not applicable (non-infectious, non-environmental).

14. Other Species / Natural Disease

- **Affected species (natural):** human (*Homo sapiens*, NCBI:txid9606) only. **No naturally occurring CHKA-equivalent disease in OMIA** for companion animals/livestock. (The spontaneous *Chkb* rmd mouse is the paralog.)
- **Orthologs:** mouse *Chka* (12660, txid10090), rat (29194, txid10116), zebrafish (558499, txid7955), *C. elegans* (180703, txid6239), plus dog/cattle/chicken/chimp (HomoloGene 88575). Deeply conserved (metazoa → yeast CKI1).
- **Comparative biology:** Kennedy pathway deeply conserved; yeast complementation of human variants demonstrates conserved catalytic function (PMID: 35202461). No zoonotic/veterinary relevance (genetic disorder).

15. Model Organisms

- **Mouse (MGI):** homozygous *Chka* KO is **embryonic lethal** (PMID: 20026284) → CHKA essentiality; a **conditional/hypomorphic or humanized knock-in** model is needed to recapitulate NEDMIMS. The *Chkb* rmd mouse models the paralogous disease and is **AAV-rescuable** (PMID: 31216357).
- **Yeast (SGD):** cell-based complementation validated reduced activity of patient variants (PMID: 35202461).
- **C. elegans (WormBase)** and **zebrafish (ZFIN):** orthologs available; no published NEDMIMS-specific model (zebrafish is a tractable CNS/microcephaly option).
- **Cellular/iPSC/organoid:** none published; **patient iPSC cortical neurons/organoids with lipidomics** is the clear next step.
- **Recapitulation:** no model fully reproduces the human CNS phenotype; existing systems establish loss of enzymatic function/essentiality and enable variant functional testing and therapeutic proof-of-concept.

Evidence Base

PMID	Title (abbrev.)	Role in this report
35202461	*Bi-allelic variants in CHKA cause a neurodevelopmental disorder with epilepsy and microcephaly*	**Defining paper.** Gene, inheritance, n=6 cohort, phenotype, LoF/Kennedy mechanism (yeast + structural). Supports Findings 1, 2, 5, 9.
27769579	*Molecular structure and differential function of choline kinases CHKα and CHKβ*	Defines choline→phosphocholine→CDP-choline→PC chain; citicoline rationale. Supports Findings 2, 12.
20026284	*Differential expression of choline kinase isoforms in skeletal muscle (rmd mouse)*	Murine *Chka* disruption embryonic lethal vs *Chkb* dystrophy; supports hypomorphic-allele inference and tissue specificity. Supports Findings 4, 8, 15.
31216357	*Functional rescue in a mouse model of megaconial muscular dystrophy*	AAV6-*Chkb* gene therapy rescues paralogous disease — therapeutic proof-of-concept. Supports Finding 4; treatment outlook.
23945283	*Megaconial CMD due to loss-of-function in choline kinase β*	Characterizes paralogous *CHKB* disorder; contextualizes disease family and differential diagnosis.
31745227	*CHKA–c-Src SH3 interaction*	Non-catalytic scaffolding role of CHKA; possible additional mechanism (unproven in NEDMIMS).
27705917 / 27206796 / 34416377	CHKA in cancer "cholinic phenotype" / inhibitor development	Provide enzymatic/structural context; illustrate that cancer biology is the mirror image (gain) of this LoF disorder.

Peripheral literature (PPFIBP1-, ACBD6-, RBL2-, BORCS5-, tubulin-, GABRB2-related disorders) was reviewed to establish the **differential-diagnostic landscape** of monogenic microcephaly–DD–epilepsy–movement-disorder syndromes; these overlap clinically but are mechanistically distinct from CHKA/Kennedy-pathway disease.

Limitations and Knowledge Gaps

1. **Extremely small evidence base.** The clinical definition rests on **one case series (n=6, 5 families)**. Prevalence, penetrance detail, natural history, survival, and genotype–phenotype correlations are essentially uncharacterized; phenotype frequencies (Finding 5) will shift as cases accrue.
2. **No patient-level molecular profiling.** No transcriptomic/proteomic/lipidomic/metabolomic patient datasets; no validated biomarker. Direct demonstration of reduced brain PC in patients is lacking (mechanism inferred from enzymology + models).
3. **No dedicated in-vivo NEDMIMS model.** Complete *Chka* KO is embryonic lethal, so a hypomorphic/conditional CNS model recapitulating the human disease is not yet available; yeast validates activity loss but cannot model neurodevelopment.
4. **Ontology/identifier gaps.** No Orphanet, ICD-10/ICD-11, or MeSH-specific code located; classification relies on OMIM/MONDO/UMLS.
5. **Therapeutics unproven.** Choline/CDP-choline supplementation and gene replacement are only theoretically motivated (by paralog biology); no experimental or clinical data in NEDMIMS.

Proposed Follow-up Experiments / Actions

1. **International case ascertainment (GeneMatcher / Matchmaker Exchange):** expand the cohort to refine phenotype frequencies, penetrance, expressivity, natural history, and genotype–phenotype correlations.
2. **Patient lipidomics/metabolomics:** measure phosphocholine, CDP-choline, and PC species in patient fibroblasts, plasma, and neural cells to establish a candidate biomarker and confirm the membrane-lipid deficit in vivo.
3. **Faithful neuronal disease model:** generate patient-derived iPSC cortical neurons/organoids or a **conditional/hypomorphic *Chka* mouse** (bypassing embryonic lethality) to test whether reduced activity causes microcephaly, hypomyelination, and hyperexcitability, and to define the developmental critical window.
4. **Structure–function mapping:** dock disease missense variants (p.Arg141Trp, p.Pro194Ser, p.Phe341Leu) onto CHKα structures (PDB 2IG7/2CKQ) and correlate predicted

destabilization/active-site perturbation with residual activity, to build a variant-effect framework for reclassifying the 54 current VUS.

5. **Therapeutic proof-of-concept:** test **choline/CDP-choline (citicoline) supplementation** and **AAV-CHKA gene delivery** in patient iPSC neurons and any hypomorphic mouse model, leveraging the AAV6-*Chkb* rmd-mouse rescue precedent (PMID: 31216357).
6. **Registry and counseling infrastructure:** establish a NEDMIMS/CHKA registry, standardized genetic-counseling/prenatal-testing pathways for affected (often consanguineous) families, and pursue Orphanet/ICD-11 cataloguing with carrier-frequency estimation.

Report compiled from 9 confirmed findings and 32 reviewed papers across 5 investigation iterations. Evidence source types: predominantly human clinical (defining case series), supported by model organism (mouse, yeast, C. elegans), in vitro (enzyme/structural), and computational (gnomAD constraint, structural modeling) data.
